

Junxiang Huang

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Education

Peking University

PhD Candidate in Computer Software and Theory

Sep. 2022 – Expected Jun. 2027

Beijing, China

Peking University

B.S. in Physics and B.S. in Computer Science

Sep. 2018 – Jul. 2022

Beijing, China

Research Experience

Xiao Yuan's Group

Center on Frontiers of Computing Studies, Peking University

Research in Quantum Information and Quantum Computing

Apr. 2021 – Present

Beijing, China

- Developed resource-efficient quantum algorithms and theoretical tools for many-body simulation and information extraction, spanning variational, tensor-network-assisted, and early fault-tolerant approaches.
- Designed qubit-, measurement-, and sample-efficient protocols for estimating local observables and nonlinear quantum-state properties, including partial-transpose moments.
- Current projects include response-transform and counterdiabatic algorithms, fault-tolerant resource estimation, and learning and simulation of open quantum dynamics.

Yuanjie Exploration

Research Intern in Quantum Error Correction

- Participating in quantum error correction research and product development.

Sep. 2026 – Present

Quantum Information Theory Group at Imperial College London

Electrical and Electronic Engineering Department, Imperial College London

Visiting Research Student

Jul. – Aug. 2023

London, UK

- Studied applications of perturbative quantum simulation to quantum information and many-body dynamics.

Research Interests

- Resource-efficient quantum algorithms for many-body simulation, eigenstate and dynamical problems, with particular emphasis on the early fault-tolerant regime.
- QEC/QEM-aware estimation protocols, algorithm–architecture resource analysis, and hybrid quantum–classical methods for quantum simulation.

[10] **J. Huang**, C. Li, L.-F. Zhang, et al. Breaking the Multiplicative Overhead in Quantum Entropy Estimation. arXiv:2609.40179, 2026.

[9] Y. Tang*, Z. Huo*, **J. Huang***, et al. Low-cost algorithm-to-execution framework for surface-code quantum computing. arXiv:2609.10965, 2026.

*Equal contribution.

[8] J. Cai*, **J. Huang***, F. Otto*, Y. Li*, et al. Generalized Mermin Inequalities for Benchmarking Large-Scale GHZ States. arXiv:2607.23574, 2026.

*Equal contribution.

[7] J. Sun*, B. Zhou*, J. Xu*, Y. Yao*, Z. Du*, Z. Zhang*, Y. Gu*, **J. Huang**, et al. Quantum-classical crossover in fault-tolerant quantum dynamics simulation. arXiv:2607.16116, 2026.

*Equal contribution.

[6] **J. Huang**, X. Wang, X. Yuan, Y. Zhang. Streaming the partial-transpose moment hierarchy with order-independent quantum memory. arXiv:2606.14204, 2026.

[5] Z. Wang*, **J. Huang***, R. Ye*, et al. A Review of Variational Quantum Algorithms: Insights into Fault-Tolerant Quantum Computing. Front. Phys. 22, 023301 (2027); online July 2026.

*Equal contribution.

[4] Q.-M. Ding, J. Peng, **J. Huang**, et al. Obtaining Accurate Ground-State Properties on Near-term Quantum Devices. Phys. Rev. Applied 25, 054057 (2026).

[3] T. Jiang*, J. Cai*, **J. Huang***, N. Zhou*, Y. Zhang, et al. One- and two-dimensional cluster states for topological phase simulation and measurement-based quantum computation. Nat. Phys. 22, 430–438 (2026).

*Equal contribution.

[2] **J. Huang**, Y. Tang, X. Yuan. Estimating Local Observables via Cluster-Level Light-Cone Decomposition. Phys. Rev. A 113, 032414 (2026).

[1] **J. Huang***, W. He*, Y. Zhang, et al. Tensor-Network-Assisted Variational Quantum Algorithm. Phys. Rev. A 108, 052407 (2023).

*Equal contribution.

Invited Talks & Seminars

Resource-Efficient Quantum Property Estimation: Partial-Transpose Moments and Quantum Fisher Information

The University of Tokyo

Oct. 2026

Estimating Local Observables via Cluster-Level Light-Cone Decomposition

Nanjing University (online)

Dec. 2025

Efficient Estimation of Global and Local Observables in Clustered Quantum Systems

Shenzhen International Quantum Academy

Jul. 2025

Referee Service

Journal referee service: Physical Review Letters, Physical Review A, npj Quantum Information, Quantum

Conference referee service: AQIS, TQC

Awards & Honors

Award for Scientific Research, 2024–2025 Academic Year

Peking University

Dec. 2025

Special Prize (Team Award), 19th “Challenge Cup”

2025 China Youth Science and Technology Innovation Challenge Track

Nov. 2025

3rd Place Award, Student Poster Presentation

AQIS 2023 Seoul

Sep. 2023

Award for Scientific Research, 2019–2020 Academic Year

Peking University

Dec. 2020

Award for Academic Excellence, 2018–2019 Academic Year

Peking University

Dec. 2019

First Prize, National Final Round; Selected for the National Training Team

34th Chinese Physics Olympiad

Nov. 2017

Selected Funded Projects

Participant, Beijing Natural Science Foundation: Youth Project on dynamical quantum phase transitions (2026–present); Interdisciplinary Key Project on practical quantum information processing (2025–present).

Patents

- Method and Apparatus for Quantum Variational Molecular Dynamics Simulation (CN119811513B)
- Hybrid Computing Method and Apparatus Based on Neural Network and Quantum Circuit (CN114742228B)

Poster Presentations

Estimating Local Observables via Cluster-Level Light-Cone Decomposition

AQIS 2026 Daejeon

Aug. 2026

Simultaneous Estimation of Partial-Transpose Moments with Active Memory Independent of the Moment Order

AQIS 2026 Daejeon

Aug. 2026

Efficient Estimation of Global and Local Observables in Clustered Quantum Systems

AQIS 2025 Hong Kong

Aug. 2025

Error Mitigation via N-Representability conditions

TQC 2024 Okinawa and SEEQA 2024 Oxford

Sep. 2024

Tensor-network assisted variational quantum algorithm

AQIS 2023 Seoul

Aug. 2023

Teaching and Mentoring Experience

Peking University (Class Advisor / Undergraduate Cohort Counselor)

EECS Class 10 (Cohort of 2022)

Sep. 2022 – Jul. 2026

Peking University (Teaching Assistant)

Quantum Information

2023 Spring

Peking University (Teaching Assistant)

Probability and Statistics (A)

2022 Spring

Specialized Skills

Languages: Chinese Mandarin (native speaker), English (fluent)

Programming Languages: Python (mostly used), C++ (familiar), Julia (beginner)